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Nursing postgraduates' innovative behaviour, influencing factors and educational requirements: a national cross-sectional study

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Abstract

Background Innovative behaviours are important for nursing postgraduates who are a significant reserve for future nursing career, thinking critically, utilizing technologies and solving complex problems. In nursing undergraduates' education, innovative behaviour is closely linked with the development of professional identity. However, there are limited studies exploring innovative behaviours, influencing factors and educational requirements of nursing postgraduates.

Aim To investigate nursing postgraduates' innovative behaviour, influencing factors, and the correlation between innovative behaviour and professional identity. Moreover, exploring nursing postgraduates' requirements for innovational education.

Methods A national descriptive cross-sectional study was conducted among postgraduate students in Chinese nursing schools between October and November 2022. A general information questionnaire, The Nurses' Innovative Behaviour Scale, the Professional Identity Questionnaire for Nurse Student, and a multiple-choice question about students' innovation education requirements were collected. The independent-sample *t*-test, one-way analysis of variance (ANOVA), multiple linear regression, and Pearson's correlation analyses were conducted.

Results This study recruited 426 nursing postgraduates from 44 nursing schools across China. The mean score of innovative behaviour of nursing postgraduates was 38.96 ± 4.90 (min-max 10–50). Nursing postgraduates' professional identity was positively correlated with innovative behaviour ($r = 0.436$, $P < 0.01$). Participants who read scientific literature frequently, had a high level of professional identity, and grew up in urban areas had high levels of innovative behaviour (adjusted $R^2 = 0.244$, $P < 0.001$).

Conclusions It was determined that the level of innovative behaviour is associated with their professional identity and the frequency of reading scientific literature in nursing postgraduates. It is imperative for educators to foster an environment that encourages postgraduate nursing students to engage with a wider array of scholarly literature and elevates postgraduate nursing students' professional identity. Furthermore, the dissemination of advanced knowledge

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and the adoption of diverse pedagogical approaches are likely to meet the innovative demands of nursing postgraduates.

Keywords Postgraduate nursing student, Education, Innovation, Professional identity

Background

The shortage of nurses is a serious global challenge [1], especially in China [2]. Notably, a significant number of nursing students express an intention to leave the nursing field even before they begin their careers [3]. Moreover, many practicing nurses tend to leave the profession due to the various pressures and challenges, further deepening the shortage [4, 5]. While the nurse shortage presents a formidable challenge, the advent of high-tech nursing devices and AI-based solutions offers promising avenues for enhancing nursing capabilities and quality of care. With rapidly advancements in medical technology, more high-tech medical devices are being applied to health care workers [6, 7]. Nursing practice transcends traditional boundaries, incorporating cutting-edge capabilities like robotics [8], virtual reality [9], artificial intelligence (AI) [10], and big data [11] to enhance quality of care and operational efficiency. By addressing underlying issues contributing to nurse attrition and embracing innovative technologies, it is possible to build a stronger, more resilient nursing workforce capable of meeting the evolving healthcare needs of society.

Higher-educated nurses are the most in-demand workforce in healthcare facilities [12]. In China, nursing undergraduate programs typically span four years, fostering foundational skills and academic interest [13]. Postgraduate studies usually last three years [14], enhancing specialized knowledge and comprehensive abilities [15]. While a bachelor's degree forms the cornerstone of professional nursing practice, advanced degrees such as master's or doctoral degrees are crucial for developing expertise in specialized areas [16, 17]. Over recent years, there has been a dramatic increase in the number of nurses pursuing higher education in China [18], underscoring the need for continuous improvement in educational quality [19]. High-quality nursing education is pivotal not only for cultivating comprehensive competence [20], but also for delivering superior health services [21], and ensuring high job satisfaction [22].

Innovative behaviour stands as a core indicator of excellence in nursing education [16]. This concept encompasses the entire lifecycle of innovation from idea generation, evaluation, obtaining support, and implementation [23, 24]. By fostering these skills, nursing education prepares professionals who can drive positive change and enhance patient care through creativity and initiative. Nurses who exhibit innovative behaviours are more likely to develop positive attitudes toward adopting innovative technologies [25]. By engaging in creative

problem-solving, these healthcare professionals foster a culture of continuous improvement, which not only enhances the overall quality of care but also improves the performance of the organization [26, 27]. The education and cultivation of innovative behaviour with postgraduate programs in China remain at an early stage of development [20]. There remains significant potential for advancing the integration of innovation methodologies into postgraduate curricula and fostering an educational environment that consistently supports creative thinking and practical application [28]. Nursing educators play a crucial role in guiding the development of students' innovative capabilities [29]; Despite significant efforts, challenges remain in cultivating key competencies among nursing postgraduates [16].

A nurse's professional identity, their self-perception regarding professional roles, and the tendency to determine professional behaviour [30], which plays an important role in facilitating the transition from nursing students into registered nurses [31]. Professional identity serves as an essential prerequisite for planning effective nursing education programs [32]. Nursing students who develop a robust professional identity are more likely to remain committed to the nursing profession after graduation [33], exhibit greater capability and confidence [34], and experience enhanced career development opportunities [30]. Nurturing a positive and flexible professional identity is essential for performing multiple roles in quality patient care [35] and reducing turnover rates within the nursing workforce [36]. Furthermore, nurses with well-developed professional identity are better positioned to prioritize patients' needs and emotions, thereby improving patients' care and safety [37].

Nursing postgraduates represent the senior reserve talent pool in nursing profession [38], making it imperative to enhance their innovative behaviour capabilities. Existing quantitative studies have demonstrated that the level of innovative behaviour among practicing nurses [39–41], postgraduate nursing students [42], and undergraduate nursing students [23, 30] are at an intermediate or higher level. These studies also indicate that educational attainment [39], and participation in innovation activities such as inventions or research projects [41] significantly influence innovative behaviour. Qualitative research has provided deeper insights into the experience of Master of Nursing Specialist (MNS) students regarding the cultivation of practical innovative abilities [16] and highlights the importance of defining individual innovation within the context of nursing practice [43]. It underscores that

postgraduate education should emphasize developing students' innovative capabilities alongside their clinical proficiency [16]. This dual focus is critical for preparing graduates to meet the multifaceted challenges posed by contemporary healthcare environments [20]. While there is evidence from undergraduates nursing education has shown that innovative behaviour is positively correlated with professional identity [30]. However, the relationship between these factors among nursing postgraduates remains underexplored.

Nursing postgraduate education plays a crucial role in preparing students to become advanced practice nurses [39]. However, many aspects of innovative behavior among nursing postgraduates remain underexplored. To address this gap and cultivate innovative nursing talent in the digital era, we conducted a national investigation to examine the current state and influencing factors of innovative behaviour among nursing postgraduates, along with exploring the relationship between innovative behaviour and professional identity. Furthermore, we delved into the specific requirements nursing postgraduates have for innovation education.

Methods

Design

This study used a cross-sectional quantitative research design along with an open-ended question, followed the STROBE guidelines (Additional File-Table S1).

Aims

The objective of this study is to investigate innovative behaviour of nursing postgraduates, influencing factors, and the correlation between innovative behaviour and professional identity. Moreover, it aims to explore educational requirements for innovation in nursing postgraduates.

Participants

Convenience sampling was used to recruit participants, and the study sampling period was from 14 October to 6 November 2022. The inclusion criteria were as follows: (i) nursing postgraduates at Chinese universities and (ii) willingness to participate in the study. Participants with reading difficulties were excluded. Completed questionnaires containing all the same options were classified as low-quality responses and excluded [44].

The sample size calculation was conducted using the formula [45]: $n = [\mu_{\alpha/2}\sigma/\delta]^2$, $\alpha = 0.05$, $\mu_{\alpha/2} = 1.96$ (critical value for a two-tailed test), $\sigma = 6.07$ (based on previous research [46]), and $\delta = 0.8$ (minimum detectable effect size). Anticipating the potential invalid samples, the final sample size was increased by 20%. Accordingly, 277 participants were required to ensure sufficient statistical power and validity of the results.

A total of 437 nursing postgraduates from 44 nursing schools and 34 cities were surveyed by utilizing questionnaires, response rate was 63.89% (437/684). 11 were excluded (all questionnaires containing the same options), and data from 426 Chinese nursing postgraduates was analysed.

Data collection

The questionnaire was drafted using Questionnaire Star (<http://www.wjx.cn>). After obtaining approval from the nursing schools, it was disseminated through WeChat groups of the nursing school faculties. Faculty members subsequently shared the link with postgraduate students' WeChat groups to ensure broad reach and participation. All the participants were informed of the purpose, content, and implications of this study before questionnaire administration via Questionnaire Star. The questionnaire was administered after informed consent was obtained.

Instruments

General information questionnaire

A general information questionnaire was administered. It included nursing postgraduates' gender, age, grade, affection for the nursing discipline, main growth environment, main growth place, participation in innovation-related courses, long-term nursing career plans (five years and above), short-term nursing career plans (less than five years), participation in publishing an article, participation in an invention, the frequency of reading scientific literature, and whether they think the innovation curriculum is necessary.

Educational requirements questionnaire

When participants think that the innovation curriculum was necessary, they were asked to choose the desired content and methods of the curriculum.

The nurses' innovative behavior scale (NIBS)

The Nurses' Innovative Behavior Scale was used to evaluate nursing postgraduates' innovative behaviours. The questionnaire comprises 10 items in three dimensions, including "ideas generation" (three items), "support obtaining" (four items), and "ideas realisation" (three items). Responses are scored on a five-point Likert scale, with the total score ranging from 10 to 50. Higher scores indicate better innovative behaviour. This scale was developed by Bao et al. in the Chinese language (Mandarin), and its Cronbach's alpha coefficient was 0.879 [47]. In this study, the scale's Cronbach's alpha coefficient was 0.898, and the Kaiser-Meyer-Olkin (KMO) was 0.918 ($P < 0.01$).

The professional identity questionnaire for nurse student (PIQNS)

PIQNS was used to evaluate nursing postgraduates' professional identity. The questionnaire comprises 17 items in five dimensions, including "career self-concept" (six items), "retention benefit and turnover risk" (four items), "social comparison and self-reflection" (three items), "career choice and autonomy" (two items), and "social persuasion" (two items). Responses are scored on a five-point Likert scale, with the total score ranging from 17 to 85. Higher scores indicate a higher level of professional identity. This scale was developed by Hao Yufang in the Chinese language (Mandarin) and its Cronbach's alpha coefficient was 0.83 [48]. In this study, the scale's Cronbach's alpha coefficient was 0.907, and the Kaiser-Meyer-Olkin (KMO) was 0.906 ($P < 0.01$).

Data analysis

Statistical analysis was conducted using SPSS 22.0 (IBM, Chicago, IL). The scores for nursing postgraduates' innovative behaviour and professional identity were normally distributed. thus, we used the independent-samples *t*-test and one-way analysis of variance (ANOVA), and Multiple linear regression analysis was utilized to explore factors influencing the total score of the innovative behaviours for nursing postgraduates. Pearson correlation analyses and multiple linear regression were also performed. Pearson correlation analysis was employed to validate the correlation between the items and the total scores of the two scales including the innovative behaviours and the professional identity for nursing postgraduates. Statistical significance was set at $P < 0.05$, and the significance level was set at $\alpha = 0.05$.

Ethical considerations

The study was approved by the Ethics Committee of the authors' university and the approval number was No. 2021(277). The research was conducted in accordance with the principles of the Declaration of Helsinki.

Results

Demographic characteristics of participants

The mean age of the investigation's 426 nursing postgraduates from 44 nursing schools in China was 25.28 ± 3.79 years old (min-max 21–46). Most were female ($n = 390$, 91.5%), from rural ($n = 214$, 50.2%) and southwest China ($n = 175$, 41.1%). 252 (59.2%) were first grade nursing postgraduates and 71 (16.7%) were third grade nursing postgraduates. 261 nursing postgraduates had taken innovation related courses (61.3%). However, over half of the nursing postgraduates did not have long-term nursing career plans ($n = 246$, 57.7%), did not participate in publishing an article ($n = 277$, 65%) or invention ($n = 346$,

81.2%). 236 (55.4%) nursing postgraduates reading scientific literature frequently (Table 1).

Innovative behaviour's score of nursing postgraduates

The mean score of innovative behaviour of nursing postgraduates was 38.96 ± 4.90 (min-max 10–50). Who like nursing discipline very much had the highest score (43.41 ± 5.58) of innovative behaviour, while the minimum score (24.00 ± 8.49) was nursing postgraduates who never read scientific literature (Table 1).

Professional identity's score of nursing postgraduates

The mean score of professional identity of nursing postgraduates was 60.15 ± 8.01 (min-max 17–85). The highest score (71.91 ± 6.41) of professional identity of nursing postgraduates belonged to "the affection for the nursing discipline who like very much", while the minimum score (49.38 ± 12.25) belonged to "the affection for the nursing discipline who dislike very much" (Table 1).

Relationship between innovative behaviour' and professional identity

Nursing postgraduates' professional identity was positively correlated with innovative behaviour ($r = 0.436$, $P < 0.01$). All dimensions of innovative behaviour were significantly positively correlated with all dimensions of professional identity ($P < 0.01$, min-max 0.144–0.513) (Table 2).

Innovative behaviour's influencing factors of nursing postgraduates

According to the requirements of multiple linear regression analysis for the inclusion of equation variables, disordered multi-categorical variables such as main growth place should be set as dummy variables. Ordinal multi-categorical variables such as affection for the nursing discipline, and frequency of reading scientific literature were assigned 1~5 points from low to high (Additional File-Table S2).

With multiple stepwise linear regression analysis, the results showed that nursing postgraduates' professional identity, main growth environment, and frequency of reading scientific literature were associated with their innovative behaviour (Table 3), which explained 24.4% (adjusted $R^2 = 0.244$) of the variance ($F = 46.621$, $P < 0.001$). And this model's Durbin-Watson (DW) test was 2.029 which means there was no autocorrelation.

Educational need of postgraduates' innovation

95.5% nursing postgraduates (407/426) reported that the innovation curriculum was necessary, and the curriculum needed plentiful content and various forms. 400 participants (98.3%) desired to know the frontier/hotspots

Table 1 General information and participants' scores on innovative behavior and profession identity ($n = 426$)

Variable	Quantity (n/%)	Innovative behaviour Mean±SD	t/F	P	Profession identity Mean±SD	t/F	P
Gender							
Female	390 (91.5)	38.96±4.87	0.02	0.984	60.14±7.83	-0.10	0.924
Male	36 (8.5)	38.94±5.31			60.28±10.50		
Age (years)							
21–25	295 (69.2)	38.88±4.96	1.58	0.194	59.79±8.07	1.82	0.144
26–30	92 (21.6)	38.98±4.93			60.16±7.82		
31–35	27 (6.3)	38.48±4.28			62.52±8.66		
36–46	12 (2.8)	41.92±4.01			63.83±8.10		
Grade							
First	252 (59.2)	39.28±5.02	3.33	0.037	60.78±7.57	2.52	0.081
Second	103 (24.2)	37.88±4.59			59.83±8.29		
Third	71 (16.7)	39.38±4.71			58.41±9.28		
Affection for the nursing discipline							
Dislike very much	13 (3.1)	38.15±5.21	9.26	<0.001	49.38±12.25	73.36	<0.001
Dislike	19 (4.5)	35.11±7.17			50.26±5.86		
Neutral	208 (48.8)	38.47±4.83			57.10±6.14		
Like	164 (38.5)	39.50±4.03			64.46±5.68		
Like very much	22 (5.2)	43.41±5.58			71.91±6.41		
Main growth environment							
Urban	212 (49.8)	39.87±5.09	3.87	<0.001	60.42±8.66	0.67	0.501
Rural	214 (50.2)	38.06±4.54			59.89±7.47		
Main growth place							
East China	51 (12.0)	40.69±5.04	2.59	0.018	61.45±9.34	1.62	0.140
South China	27 (6.3)	39.07±3.93			59.89±5.11		
Central China	85 (20.0)	37.55±4.55			59.55±8.19		
North China	23 (5.4)	38.17±4.92			57.43±6.42		
Northwest China	27 (6.3)	39.52±5.37			57.30±8.88		
Southwest China	175 (41.1)	38.98±4.89			60.58±7.57		
Northeast China	38 (8.9)	39.71±5.14			61.68±9.73		
Participated in innovation-related course							
Yes	261 (61.3)	39.42±4.85	2.46	0.014	60.65±8.14	1.60	0.111
No	165 (38.7)	38.23±4.91			59.37±7.94		
Having long-term nursing career plans							
Yes	180 (42.3)	40.11±5.30	4.21	<0.001	62.97±8.29	6.45	<0.001
No	246 (57.7)	38.12±4.41			58.09±7.84		
Having short-term nursing career plans							
Yes	333 (78.2)	39.56±4.73	4.90	<0.001	61.70±7.64	8.02	<0.001
No	93 (21.8)	36.82±4.92			54.61±7.16		
Participation in publishing an article							
Yes	149 (35.0)	39.49±5.36	1.64	0.102	59.06±8.79	-2.06	0.040
No	277 (65.0)	38.68±4.62			60.74±7.62		
Participation in an invention							
Yes	80 (18.8)	39.59±5.67	1.13	0.260	60.33±9.06	0.21	0.835
No	346 (81.2)	38.82±4.70			60.12±7.85		
Frequency of reading scientific literature							
Never	2 (0.5)	24.00±8.49	9.52	<0.001	52.50±3.54	1.46	0.214
Rarely	64 (15.0)	37.64±4.73			58.80±9.01		
Sometimes	124 (29.1)	38.37±4.71			59.77±7.84		
Often	200 (46.9)	39.49±4.51			60.64±8.08		
Always	36 (8.5)	41.25±5.61			61.64±6.88		
Innovation curriculum is necessary							

Table 1 (continued)

Variable	Quantity (n/%)	Innovative behaviour Mean ± SD	t/F	P	Profession identity Mean ± SD	t/F	P
Yes	407 (95.5)	39.09 ± 4.76	2.62	0.009	60.34 ± 7.69	2.22	0.027
No	19 (4.5)	36.11 ± 6.87			56.16 ± 13.71		
Total scores		38.96 ± 4.90			60.15 ± 8.01		

Table 2 Correlation between innovative behavior and professional identity (n = 426)

Variable	1	2	3	4	5	6	7	8	9	10
1. Innovative Behaviour	1									
2.IB-SUB1	0.786**	1								
3.IB-SUB2	0.929**	0.605**	1							
4.IB-SUB3	0.878**	0.492**	0.772**	1						
5. Professional Identity	0.436**	0.390**	0.372**	0.379**	1					
6.PI-SUB1	0.333**	0.271**	0.290**	0.305**	0.933**	1				
7.PI-SUB2	0.296**	0.254**	0.250**	0.268**	0.893**	0.851**	1			
8.PI-SUB3	0.510**	0.513**	0.428**	0.400**	0.583**	0.389**	0.347**	1		
9.PI-SUB4	0.167**	0.146**	0.144**	0.145**	0.360**	0.204**	0.235**	0.118**	1	
10.PI-SUB5	0.352**	0.342**	0.291**	0.292**	0.660**	0.516**	0.444**	0.420**	0.155**	1

Note: IB: innovative behaviour; IB-SUB1: ideas generation; IB-SUB2: support obtaining; IB-SUB3: ideas realization; PI: professional identity; PI-SUB1: career self-concept; PI-SUB2: retention benefit and turnover risk; PI-SUB3: social comparison and self-reflection; PI-SUB4: career choice and autonomy; PI-SUB5: social persuasion. **: $P < 0.01$

Table 3 Multiple linear regression analysis for factors influencing nursing postgraduates' innovative behavior

Item	B	95%CI		S.E.	Beta	t	P	VIF
		Lower	Upper					
Constant	22.832	19.221	26.444	1.837		12.426	< 0.001	
Profession identity	0.250	0.199	0.301	0.026	0.412	9.709	< 0.001	1.012
Main growth environment	-1.551	-2.365	-0.737	0.414	-0.158	-3.745	< 0.001	1.006
Frequency of reading scientific literature	0.984	0.511	1.457	0.241	0.174	4.086	< 0.001	1.017

Note: B is the unstandardised regression coefficient, CI is the confidence coefficient, S.E. is the standard error, and Beta is the standardised regression coefficient. VIF is the variance inflation factor

Table 4 Innovation education content and forms of participants' needs (n = 407)

Variable	Quantity (n/%)
Content	Innovation theory education 299 (73.5%)
	Innovation methods education 347 (85.3%)
	Innovation thinking education 362 (88.9%)
	The frontier/hotspots in nursing 400 (98.3%)
	Innovative ability development 346 (85.0%)
Methods	Traditional theory teaching 303 (74.4%)
	Problem-based learning (PBL) 278 (68.3%)
	Team-based learning (TBL) 309 (75.9%)
	Expert lectures 336 (82.6%)
	Diversity teaching 359 (88.2%)

in nursing, 359 participants (88.2%) want have diversity teaching methods (Table 4).

Discussion

The findings of this study reveal that nursing postgraduates exhibit a generally high level of innovative behaviour. These students are actively involved in critical thinking, information-seeking, creative problem-solving, and the adoption of novel approaches within their academic and

clinical practice [49]. A positive attitude towards innovation is essential for advancing nursing practices and enhancing patient care outcomes [41]. Furthermore, the study found that the frequency of reading scientific literature, professional identity, and the primary growth environment are significantly related to innovative behaviours among nursing postgraduates.

The scores of nursing postgraduates' innovative behaviour in our study were higher than those of clinical nurses [41, 50], undergraduate nursing students [30], and were comparable to findings from previous studies on nursing postgraduates [51]. This difference can be attributed to several key factors. In China, undergraduate nursing students undergo extensive practical training intensity, including over ten months of clinical internships during their final year [52], alongside completing a bachelor's thesis [13]. This rigorous combination of practical and research activities helps cultivate both clinical skills and academic capabilities. Before admission into a postgraduate nursing program, applicants must undergo an interview process designed to evaluate their potential for research and scientific inquiry. This assessment helps

ensure that incoming students possess the necessary qualities and aptitudes to excel in advanced academic and research environments [53, 54]. Consequently, those who advance to the graduate level are typically well-prepared with robust clinical and research experience, setting a strong foundation for further academic and professional development. Additionally, postgraduate nursing students are more deeply involved in academic learning compared to clinical nurses, who often face significant time constraints and heavy workloads [55]. This deeper engagement fosters a greater inclination towards innovative practices and scientific inquiry, reflecting their advanced training and educational background. These factors collectively contribute to the higher levels of innovative behaviour observed in our study.

Frequent engagement with scientific literature is associated with higher levels of innovative behaviour among nursing postgraduates. Mitchell's study had found that students' pre-admission reading ability can contribute to academic success [56], where "reading ability" primarily refers to the skills required for reading comprehension in courses or tests. In our study, we found that the frequency of reading scientific literature positively correlates with higher innovative behaviour, aiding students in more effectively solving academic and career-related problems [26]. Through regular exposure to scientific literature, nursing postgraduates can stay abreast of cutting-edge advancements in the medical field and broaden their horizons. To facilitate this process, initiatives like Nursing Journal Club or Virtual Nursing Journal Club can be valuable resources for helping students obtain and engage with scientific literature [57]. To maximize the effectiveness of literature reading, postgraduates should also develop skills in analyzing, comprehending, and critically thinking about the content of scientific articles [56].

The professional identity of nursing postgraduates was found to be positively correlated with their innovative behaviour, consistent with findings from previous studies [30, 58]. A strong professional identity positively influences the core competencies of nursing students, including their innovation ability [59]. Moreover, professional identity can enhance innovative behaviour by increasing the motivation [60], which in turn encourages students to actively acquire knowledge [61]. Additionally, there is a reciprocal relationship between professional identity and innovative behaviour. Not only does a high professional identity foster greater innovative behaviour, but engaging in innovative activities also strengthens professional identity [30]. This positive feedback loop can simultaneously improve both aspects, helping nursing postgraduates' adaptation to the rapidly evolving better field of digital healthcare.

The professional identity of nursing graduate students in this study was found to be at a middle level, consistent

with previous studies on nursing students [62, 63]. Professional identity is a dynamic process that evolves over time [64], particularly through experience gained in clinical settings where theory and practice can be integrated [65].

Students who have career plans tend to score higher in professional identity. Career planning helps nursing postgraduates choose more suitable career opportunities [66], increases their learning engagement, and promotes the development of a positive professional identity [67]. Consistent with prior research, our study also found that nursing postgraduates with a strong interest in nursing exhibited a higher level of professional identity [68]. Inner beliefs and personal passion for nursing are key factors that motivate nursing postgraduates to remain committed to their career [69], leading to a stronger sense of professional identity.

For nursing postgraduates who grew up primarily in urban areas, their innovative behaviour was notably higher. This may be attributed to greater exposure to innovative concepts and technologies, suggesting that these postgraduates can benefit from additional inspiration and support for innovation. Similarly, nursing postgraduates who grew up primarily in East China exhibited higher levels of innovative behaviour compared to those from other regions. Factors such as environment, university resources, and family support can collectively form a robust network that fosters innovation [43]. East China, being a coastal area with developed economy and advanced technology, offers nursing postgraduates more opportunities for international exchanges and scientific research. The region's economic prosperity and technological advancements likely contribute to a richer environment for fostering innovative thinking. As technology continues to advance, various methods can promote the simultaneous development of innovative behaviour across different regions. For instance, online courses and Massive Open Courses (MOOCs) are effective ways [70], that transcend time and space limitations, offering flexibility and accessibility [71]. These platforms can provide equal opportunities for nursing postgraduates from diverse backgrounds to engage in innovative activities and enhance their capabilities. These findings underscore the importance of leveraging regional advantages and technological advancements to foster innovation among nursing postgraduates. By providing targeted support and leveraging modern educational platforms like MOOCs, institutions can help bridge the gap in innovative behaviour across different regions, ensuring that all postgraduates have access to the resources they need to thrive in an increasingly innovative healthcare landscape.

In our study, the independent-sample *t*-test revealed that nursing postgraduates with career plans exhibit higher levels of innovative behaviour. Having clear plans

helps postgraduates pursue career opportunities [66], and such planning may stimulate more innovative ideas during their academic activities [72]. Moreover, our finding indicates that while 78.2% of the participants had a short-term career plan, only 42.3% had a long-term career plan, suggesting that it is challenging for postgraduates to formulate, especially long-term, career plans [52]. This difficulty underscores the importance of implementation planning in promoting innovation [73]. However, the lack of professional guidance from educators can hinder effective plan implementation [16]. Therefore, nursing educators should play a crucial role in guiding nursing postgraduates on how to develop both short-term and long-term career plans and assist them in implementing these plans. By providing structured support and mentorship, educators can help students overcome challenges and foster a more innovative and goal-oriented approach to their careers. These findings highlight the need for integrated career planning and innovation support within nursing education programs. By fostering a supportive environment where students receive guidance on career planning and innovation, institutions can better prepare future nursing professionals to meet the evolving demands of healthcare.

The influencing factors of innovative behavior, as identified through multiple linear regression may be influenced by various reasons. In the investigation, while the degree to which nursing postgraduates enjoy their nursing field and whether they perceive the innovation curriculum as necessary can influence their innovative behaviour, these factors were not included among the multiple predictors in the regression model. The two factors, however, represent important inner motivation that can help postgraduates develop innovative ability [60]. Meanwhile, students with a strong interest in self-learning tend to exhibit a higher level of engagement in learning [74]. According to the Theory of Planned Behaviour, behaviour is influenced by three main components: attitude, subjective norms, and perceived behavioural control [75]. Subjective norms refer to extrinsic factors, such as social pressure [76], which can significantly impact behaviour. Although nursing postgraduates generally have a positive attitude toward innovative behaviour in nursing, external factors (subjective norms) and perceived behavioural control can pose challenges. For instance, heavy clinical work pressure [28], a lack of scientific research capacity [16], and lower levels of creative self-efficacy [41] can hinder innovative behaviour despite a positive attitude. These findings highlight the complex interplay between internal and external factors in shaping innovative behavior among nursing postgraduates. While fostering a positive attitude towards innovation is crucial, it is equally important to address external barriers such as heavy clinical workloads and limited research

opportunities. Institutions should focus on providing adequate support and resources to enhance postgraduates perceived behavioral control, thereby promoting a more conducive environment for innovation.

Moreover, while most participants believed that an innovation curriculum is necessary, only a small proportion participated in innovation outcomes such as publishing articles or developing inventions. This suggests that there is a significant gap between recognizing the importance of innovation and translating it into tangible results [77]. Therefore, while encouraging innovative behavior among nursing postgraduates, it is crucial to provide more comprehensive support for innovation.

Educators should share the cutting-edge research and emerging trends in nursing to postgraduates. In the conventional educational paradigm, instructors typically disseminate content derived from textbooks, which, due to the publication timelines, often results in pedagogical material that is outdated related to the most recent advancements and the evolving societal demands [16]. To address this gap, postgraduates should be encouraged to read updated publications in medical field to keep abreast of the latest developments in nursing. Moreover, educators need develop their teaching guidance abilities [78] and provide more suitable environments and teaching methods to postgraduates [79]. A variety of teaching methods [80] are essential for an innovation curriculum to attract postgraduates' interests and achieve better teaching outcomes [81]. Effective teaching methods include: the creative arts [82], inquiry-oriented teaching [83], and virtual reality [84]. Furthermore, innovative education should emphasize the integration of theory with practice, as most innovative ideas stem from practical experience [85]. Regardless of the method chosen, the primary goal should not be to increase postgraduates' learning burden, but to help them improve their innovative ability and prepare effectively for their careers [16]. These strategies underscore the importance of adapting teaching practices to meet the dynamic needs of modern nursing education. By integrating up-to-date research and innovative teaching methods, educators can create a stimulating and supportive learning environment that nurtures students' innovative capabilities. This approach not only enhances students' readiness for professional challenges but also fosters a culture of continuous learning and improvement within the nursing community.

In our study, we didn't distinguish between full-time and part-time postgraduate students. Part-time nursing postgraduates often face significant challenges, including balancing daily work responsibilities with family and study commitments [86]. Consequently, they require more support from both hospitals and educational institutions to manage these competing demands effectively. Full-time nursing postgraduates, on the other hand, tend

to focus on discovering new problems and advancing theoretical knowledge, while part-time nursing postgraduates are more oriented toward solving practical clinical problems [16]. These differing priorities suggest that innovative educational requirements may vary between the two groups. Future research should investigate these differences to better understand and address the targeted educational needs of each group, thereby enhancing innovation in nursing education. These findings highlight the need for tailored educational strategies that account for the unique circumstances of full-time and part-time nursing postgraduates. By providing customized support and resources, educational institutions can better prepare postgraduates to meet the diverse challenges of modern healthcare. For part-time students, this might include flexible scheduling, mentorship programs, and practical problem-solving workshops. For full-time students, it could involve fostering a research-oriented environment with opportunities for advanced theoretical exploration and innovation.

Nursing postgraduates who possess robust innovative behavioural abilities are better positioned to adapt to the evolving demands of their environment and promote personal and professional growth [16, 20]. Our study's findings indicate that nurse educators should consider the unique circumstances and needs of each postgraduate, providing them with more personalized and comprehensive innovation education. Nurse educators play a crucial role in this process by offering support, resources, and mentorship that empower nursing postgraduates to develop and apply innovative solutions effectively.

Limitations

Our study did not distinguish between full-time and part-time postgraduate students, and students in different academic programs may have different needs. Future studies should distinguish between different academic programs, as this approach can uncover program-specific factors that influence innovative behavior. By doing so, researchers can identify unique challenges and opportunities within each program, leading to more tailored and effective educational strategies. Additionally, further research could conduct in-depth interviews regarding nursing students' experiences, which could offer richer insights.

Conclusion

The postgraduate nursing students' innovative behaviour was good but still can be improved. It is important to instruct nursing postgraduates on how to develop a career plan and help them implement it. Nursing educators are encouraged to facilitate postgraduate students' reading of contemporary scholarly articles and to enhance students' professional identity, while

nursing educators should also pay more attention to rural students. In innovation education, theory should be integrated with cutting-edge issues and practical applications. Moreover, nurse educators should employ various teaching methods to teach students, such as MOOCs, TBL, and PBL, help students move from learning to output, and promote innovative curricula that address students' needs without increasing the burden on students.

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

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Author contributions

JYZ: Formal Analysis, Statistic analysis, Writing—Original Draft Preparation. SY: Investigation, Writing—Original Draft Preparation. LHP: Writing—Review and Editing. MPR: Writing—Review and Editing. LQW: Investigation. MYQ: Writing—Review and Editing. FYZ: Methodology, Writing—Review and Editing, funding acquisition.

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Data availability

Data used to support the findings of this study are available from the corresponding author upon request.

Declarations

Ethics approval and consent to participate

The study was approved by the Ethics Committee of the authors' university and the approval number was No. 2021(277). The research was conducted in accordance with the principles of the Declaration of Helsinki. The questionnaire was administered after informed consent was obtained from participants.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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